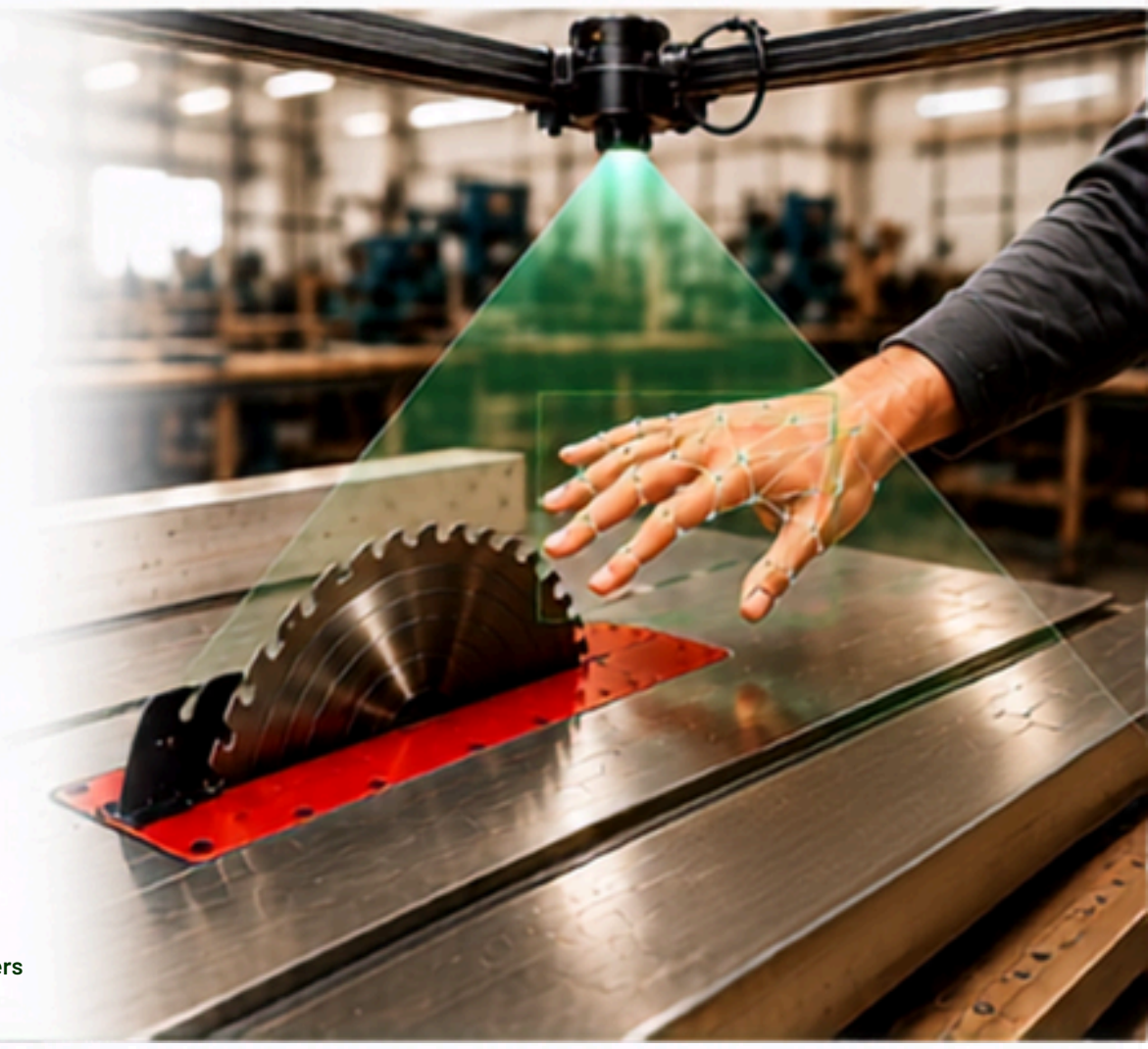
HELIOTEC
台中市大雅區雅潭路三段400巷117號 NO177, LN.400, SEC.3, YATAN RD, TAYA DIST, TAICHUNG CITY, TAIWAN, R.O.C.
TEL/04-2569-2078 FAX/04-2568-9581 meta.ruby@msa.hinet.net

AI VISION GUARDIAN

VISION-BASED HAND TRACKING & MACHINE SAFETY SYSTEM

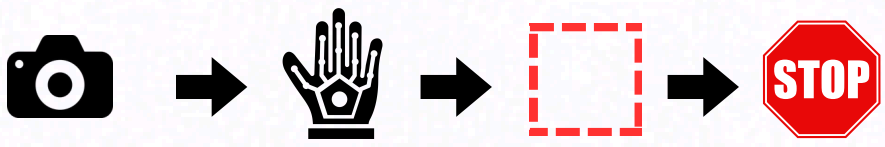
Detect. Prevent. Protect. Prevent Accidents Before They Happen.

SafeHands is an AI-powered vision system that detects hands in real time and stops the machine when they enter the danger zone. Designed to dramatically reduce hand injuries and improve workplace safety across a wide range of industrial machines.



- Real-Time Hand Tracking
- AI-Based Recognition
- Millisecond Response
- Reduce Injury Risk
- Protect Workers & Machines

HOW IT WORKS



1. Capture
Camera captures real-time video above the machine.

2. Detect
AI detects hand landmarks and position.

3. Evaluate
System evaluates if hand is in the warning or danger zone.

4. Stop
If too close, machine stops immediately via safety signal.

KEY FEATURES

- Real-Time Hand Tracking
- AI-Based Recognition (MediaPipe)
- Millisecond Response (<20ms)
- Configurable Safety Zones
- Instant Machine Stop via I/O
- Easy Integration
- Supports Multiple Machines & Custom Sets

SYSTEM TARGETS

- Response Time < 0.2 of (Blade/Motor Stop)
- Detection Range (Red Zone 0.5 cm)
- Edge AI Performance 100 TOPS
- High Accuracy 99%

IDEAL FOR MULTIPLE MACHINES


TABLE SAW


BAND SAW


CUTTING MACHINES


PRESS MACHINES


COLLABORATIVE ROBOTS


INDUSTRIAL ROLLER MILLS

Suitable for various woodworking, metalworking, and manufacturing equipment.

SYSTEM COMPONENTS



INDUSTRIAL CAMERA
See3CAM_CU30
3.4 MP onsemi AR0330
Fixed Focus
Rolling shutter
Low light performance
On-board high performance ISP



EDGE AI UNIT
NVIDIA jetson orin nano
100 TOPS Performance
8GB RAM



SIGNAL LIGHT
Green/Yellow/Red
Connected to I/O
for Safety Feedback



CAMERA MOUNT & HOLDER
Adjustable for different heights and table mounting



7" DISPLAY
Visualization of safety zones and system outputs via HDMI



QR CODES
QR codes for calibration and customization of safety zones

THREE SAFETY ZONES



SAFE ZONES
Normal Operation

WARNING ZONE
Caution - hand getting close

DANGER ZONE
Immediate stop if hand enters

MAXIMIZE SAFETY
Reduce human injuries to near zero.

LOWER COSTS
Reduce maintenance and accident-related costs.

MINIMIZE DOWNTIME
Prevent accidents and restart machines quickly.

INTELLIGENT DETECTION
AI distinguishes hands from other objects, reducing false alarms.

BOOST PRODUCTIVITY
Safer workplace leads to higher efficiency and output.

AI VISION FOR A SAFER TOMORROW.